CONSOLIDATED ENGINEERING RECORD / PUBLIC EDITION

Cross-Architecture Physical Execution, Decision Parity and Latency Evidence

FARO / OSH — ESP32-S3 · STM32F446RE ARM Cortex-M4 · CH32V307 RISC-V

30	3	49,920 B
Common reference cases	Physical processor families	Identical compiled reference table in four archival ELFs

Executive finding

Three physical microcontroller families reported passing 30-case structural-parity gates. A subsequent owner-run, read-only ELF audit identified four original binary images whose compiled 30-case reference tables are identical byte for byte. This connects the archived test design more strongly than source-file similarity alone.

Recorded route latencies exist for all three families, but they use different firmware and timing boundaries. They are reported as historical measurements, not as a synchronized cross-platform speed ranking.

This document integrates the historical physical evidence, the original 30 printed ESP32-S3 UART records, the compiled-table audit, firmware provenance and declared limitations. No new physical experiment was run to prepare it.

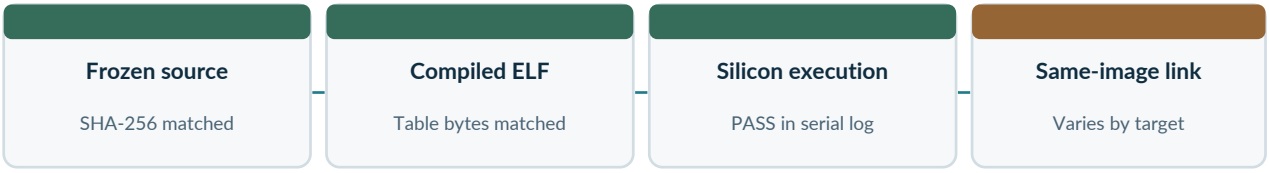
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01 / Executive conclusion and evidence hierarchy

What the records establish. Native FARO/OSH test code was physically executed on ESP32-S3, STM32F446RE and CH32V307, and each platform reported a successful 30-case structural-parity gate. The 30-case test is separate from each platform's later 100,000-cycle endurance and electrical experiments. [S1–S6]

What the 9 November audit adds. The operator locally identified the ARM parity, ARM 100K, RISC-V V2 and ESP32-S3 final ELF files by their full SHA-256 values. In each, the symbol `OSH_PARITY_CASES` resolved to 49,920 bytes across 30 records. All four extracted tables shared the same digest. The report reproduces the operator-supplied result; it was not independently rerun by an outside laboratory. [S7–S9]



A shared compiled table is not, by itself, a full historical Flash-image attestation.

Evidence types must not be conflated

Evidence level	What is demonstrated	Limit
Physical observation	Hardware executed the stated qualification; preserved serial PASS and other bench measurements.	Does not, by itself, identify every later archived file byte.
Compiled reference identity	Four SHA-256-identified ELF files contain byte-identical 30-case tables.	Does not, by itself, prove that each exact image was resident on the hardware during the historical capture.
Logical per-case inference	If the same verifier/corpus were executed, aggregate 30/30 success entails all five checks passed on every one of those 30 cases.	Not a substitute for 90 separately printed UART output lines.
Timing	Recorded internal route means and external oscilloscope intervals.	Measurement boundaries are not common across devices.

Evidence claim: physical structural parity with a byte-identical archived 30-case reference corpus. This is not a new three-chip rerun or an independent on-device Flash readback.

02 / Methodology: 30 cases, five causal boundaries

FARO is an embedded artificial-intelligence architecture, not merely a PID-like controller. This protocol evaluates a specified native state-to-selection/meaning pathway, not the complete cognitive system or all future deployments. [S1, S7]

Gate	Boundary	Observed condition required for PASS
L1	Endogenous final state	Numeric agreement against the frozen reference state.
L2	Settle and vortex trajectory	Path continuity, native-state evolution and omega agreement within declared tolerances.
L3	Homeostatic Attention Loop (HAL)	Expected candidate counts and selected indices; enforcement of authority boundary.
L4	Selector identity	Exact identity and ordered index of selected edges.
L5	Final semantic result	Expected final slots, fact hashes and final-meaning digest.

Acceptance logic

The examined verifier returns `run_one = L1 && L2 && L3 && L4 && L5`. The aggregate routine processes the 30 reference records and succeeds only when all 30 per-case predicates are true. Consequently, **for the exact verifier and compiled corpus that were executed**, an aggregate 30/30 PASS logically entails success on every one of the 30 cases. Printing an individual UART line is not a prerequisite for that logical implication. [S7–S9]

Numerical precision is an acceptance criterion, not a measured error distribution

Check	Threshold / condition	Interpretation
Strict state	2×10^{-5}	Declared strict state comparison.
Conditioned state	2×10^{-3}	Only for explicitly ill-conditioned trajectory cases.
Omega	2×10^{-5}	Declared trajectory angular/tangent-related comparison.
Selection / meaning	Exact identities / digests	Discrete equality; not bit-exact floating-point arithmetic.

Three different meanings of parity

PC-to-MCU: a board is checked against the frozen PC-derived reference. **Within-target structural:** all five gates pass on one platform. **Cross-target:** the board images use the same corpus and compatible checking logic and satisfy that common reference. The third claim additionally requires executed-image provenance; separate PASS summaries alone would not prove a common corpus. [S7]

03 / Compiled reference corpus: four-file identity audit

A local, owner-run audit on 9 November scanned 54 ELF candidates, selected four by exact full-file SHA-256 and extracted `OSH_PARITY_CASES` using ELF symbol-to-section addressing. Every selected table was 49,920 bytes, equating to 30 × 1,664-byte records, and their table digests were identical. [S9]

Role	ELF identity	Full-file SHA-256
ARM parity / 20K	`osh_stm32_uart.elf`	9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645
ARM 100K	`osh_stm32_100k.elf`	98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1
RISC-V V2 100K	`OSH_CH32V307_100K_V2.elf`	cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517
ESP32-S3 final	Final physical ELF	30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3

COMMON COMPILED TABLE / SHA-256

`4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf`

COMMON ARCHIVED SOURCE FIXTURE / SHA-256

`23863039f2b2b920fe578c6dc6e0e00624ba34e6add00709522be8d6e70ec321`

The two digests differ by design: one identifies the header/source fixture, the other identifies the initialized data embedded in linked ELF files. The frozen ARM source fixture was also re-hashed locally on 9 November and matched the reference value. [S7–S9]

Interpretation

This closes the previously outstanding **archived compiled-corpus comparison** for the four identified ELF files. It does **not** claim that machine code, peripheral libraries or startup code are identical across ISAs, nor that a table match alone proves use by a particular historical programming session.

The original checker also reports matching whole-file ELF digests, providing an unambiguous identity of the four archived artifacts, rather than selecting same-named but different firmware revisions. [S9]

04 / ESP32-S3: R2 UART, RAM execution and final campaign

The preserved ESP32-S3 R2 serial transcript records its own firmware image SHA-256, a successful image-format check, an `esptool` RAM load of four segments, followed by execution and 30 individually printed `CASE=01` through `CASE=30` records. Every printed line reports L1–L5 PASS. This is physical MCU execution rather than a host-run emulation. The R2 boot transcript reports a CPU frequency of 160 MHz; that reading is not automatically assigned to every later ESP32 firmware revision. [S1, S2]

R2 record / output	Observed evidence
Image BIN digest	`4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811`
Execution mechanism	Physical ESP32-S3; `load_ram`; all segments transferred; device executes loaded code; UART captures parity results.
Per-case result	30 original printed case lines, each L1=L2=L3=L4=L5=PASS, with NUM, HAL, SEL, FINAL and MEANING fields.
Aggregate result	Five reference levels pass; combined 150/150 checks for Step19.
Distinct ELF in compiled audit	Final physical ELF SHA-256 `30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3`.

Important identity distinction

The **R2 RAM-loaded image** that printed the 30 case records and the separately named **final physical ELF** used in the four-ELF corpus audit are **not assumed to be the same executable**. The compiled audit proves that the final ELF has the shared table; the original R2 transcript independently shows per-case results. A direct R2-ELF-to-RAM-load identity comparison is a separate provenance link.

Later 100,000-cycle physical closed loop

The 21 September dossier documents physical reference input on GPIO3, OSH processing, PWM output on GPIO14 through an external 1 kΩ / 100 µF RC network and ADC feedback on GPIO2. It recorded 100,000 cycles, INC 5,160 / HOLD 93,358 / DEC 1,482, ADC failures 0, state failures 0, a physical gate PASS and final latch PASS. Internal mean route time was 29.525 µs (24–30 µs range). These actuator totals are **not** the 30 structural-reference decisions. [S1]

05 / STM32F446RE: parity/20K and separate 100K evidence

The archived ARM parity/20K capture reports `PARITY_CASES=30`, `L1_L2_L3_L4_L5_ALL_30_CASES=PASS` and `PC_ESP32_PARITY_GATE=PASS`. Its complete SHA-256 was recomputed locally on 9 November and matched the earlier archive. [S3, S8]

ARM artifact	Reference / interpretation
Parity/20K ELF	`osh_stm32_uart.elf` · `9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645`
Parity/20K BIN	`336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae`
Parity UART capture	`c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720` (locally reverified).
ARM 100K ELF	`osh_stm32_100k.elf` · `98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1`
ARM 100K BIN	`7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997`

What was verified on the actual target

The original ARM migration record identifies `osh_stm32_100k.elf` as the programmed firmware and documents STM32CubeProgrammer's **Download verified successfully**, reset and subsequent 100K execution. It describes 30/30 parity PASS, 100,000 closed-loop cycles, 0 ADC/state failures, 36 / 99,934 / 30 INC/HOLD/DEC and a 24.937 s run. The physical loop connects DAC PA4/A2 to ADC PA0/A0. [S3, S4]

The separate programmer log naming `osh_arm_scope.elf` is a different instrumentation image and must **not** be presented as proof that the parity/20K or 100K ELF was programmed. The operator's later 100K ELF hash also does not constitute a contemporaneous on-device Flash readback. [S4, S8]

Measurement boundary

Internal route timing: 209.500 µs minimum, 241.250 µs mean, 241.875 µs maximum, using a Cortex-M cycle counter under a declared 16 MHz timing conversion (minimum 3,352, mean 3,860, maximum 3,870 cycles). A separate oscilloscope capture recorded an approximately 226 µs instrumented interval. End-of-run reference 2,662 and ADC 2,343 imply a real 319-count residual; structural PASS is not a claim of zero tracking error. [S3]

06 / CH32V307 RISC-V: V2 physical execution and separate runs

On 8 October the CH32V307VCT6 / QingKe V4F RISC-V, reporting 96 MHz target clock, performed native 30-case structural parity followed by a 100,000-cycle qualification. The preserved 11:08–11:09 timestamped capture identifies the platform, records `STRUCTURAL_PARITY_30_30=PASS`, then 100,000 completed cycles with zero state failures and zero invalid actions. [S5]

Artifact / session	Verified or preserved evidence
V2 ELF SHA-256	`cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517`
V2 HEX SHA-256	`544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21`
V2 11:08 session	30/30 PASS; 100,000 completed cycles; actions INC 174 / HOLD 99,667 / DEC 159; state failures 0; invalid actions 0.
Earlier 00:16 session	Different timestamped UART recording; archived digest `6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d`.

Identity boundary

The historical file-selection note identifies the original V2 `FLASH_100K.hex` and the matching full digest; three archived copies share it. However, the exact same-session WCHISPTool selection/verify transcript and an on-device Flash readback for the 11:08 recovery session were not recovered. This is a **historical image-attestation gap**, not a failed parity run. [S5, S9]

Do not merge distinct RISC-V experiments

The later 9 October physical DAC-to-ADC campaign used a distinct firmware HEX (digest beginning `2db7f4db`). Its actions were 36 / 99,934 / 30, not the V2 histogram above. It supplied the 356.843 µs TIM2 mean and the 284 µs externally marked wrapper interval. These measurements belong to the 9 October workload and cannot silently be assigned to V2. [S6]

The 8 October V3 DAC-to-ADC attempt had no execution UART and was explicitly marked **NOT VALIDATED** in its checkpoint. It contributes no physical PASS assertion. No cause is inferred for that missing UART. [S5]

07 / Thirty-case reference-to-target matrix

D01–D11 identify anonymized reference decision signatures, not analog actuator commands. U = printed result in the original ESP32-S3 R2 capture; G = result implied by an aggregate 30/30 gate if the executed verifier and corpus match.

Case	Class	ESP32-S3 R2	ARM	RISC-V V2
01	D01	**PASS / U**	PASS / G	PASS / G
02	D02	**PASS / U**	PASS / G	PASS / G
03	D02	**PASS / U**	PASS / G	PASS / G
04	D03	**PASS / U**	PASS / G	PASS / G
05	D03	**PASS / U**	PASS / G	PASS / G
06	D03	**PASS / U**	PASS / G	PASS / G
07	D04	**PASS / U**	PASS / G	PASS / G
08	D04	**PASS / U**	PASS / G	PASS / G
09	D04	**PASS / U**	PASS / G	PASS / G
10	D05	**PASS / U**	PASS / G	PASS / G
11	D05	**PASS / U**	PASS / G	PASS / G
12	D05	**PASS / U**	PASS / G	PASS / G
13	D06	**PASS / U**	PASS / G	PASS / G
14	D06	**PASS / U**	PASS / G	PASS / G
15	D06	**PASS / U**	PASS / G	PASS / G
16	D07	**PASS / U**	PASS / G	PASS / G
17	D07	**PASS / U**	PASS / G	PASS / G
18	D07	**PASS / U**	PASS / G	PASS / G
19	D08	**PASS / U**	PASS / G	PASS / G
20	D08	**PASS / U**	PASS / G	PASS / G
21	D08	**PASS / U**	PASS / G	PASS / G
22	D09	**PASS / U**	PASS / G	PASS / G
23	D09	**PASS / U**	PASS / G	PASS / G
24	D09	**PASS / U**	PASS / G	PASS / G
25	D10	**PASS / U**	PASS / G	PASS / G
26	D10	**PASS / U**	PASS / G	PASS / G
27	D10	**PASS / U**	PASS / G	PASS / G
28	D11	**PASS / U**	PASS / G	PASS / G
29	D11	**PASS / U**	PASS / G	PASS / G
30	D11	**PASS / U**	PASS / G	PASS / G

Matrix interpretation. The source and ELF audits establish the common archived reference corpus. The ARM/RISC-V per-case entries are logical consequences of the reported passing aggregate gates, conditional on exact execution of the reviewed verifier and corpus; they are not individually printed UART lines. Matched per-case latency was not recorded. [S7–S9]

08 / Physical endurance and hardware I/O evidence

Test campaign	Structural parity	100K output regimes	Fault summary
ESP32-S3 final physical	30 cases × 5 levels; 150/150 PASS	INC 5,160 / HOLD 93,358 / DEC 1,482	ADC=0; state=0
STM32F446RE 100K	30/30 PASS	INC 36 / HOLD 99,934 / DEC 30	ADC=0; state=0
CH32V307 V2, 8 Oct	30/30 PASS	INC 174 / HOLD 99,667 / DEC 159	state=0; invalid actions=0
CH32V307 Diligence R1, 9 Oct	30/30 PASS (separate image)	INC 36 / HOLD 99,934 / DEC 30	ADC/state failures=0

Physical input / decision / output / feedback configurations

- **ESP32-S3:** physical B25K reference → GPIO3 ADC → native FARO path → GPIO14 PWM → external 1 kΩ / 100 µF RC → GPIO2 ADC feedback. Final physical gate and latch PASS.

- **STM32F446RE:** DAC A2/PA4 → external electrical connection → ADC A0/PA0 → native state/decision loop. 100K run finished in 24.937 s.

- **CH32V307 V2 (8 Oct):** native reference-based structural parity and deterministic 100K qualification; that checkpoint did not establish a successful physical DAC-to-ADC feedback gate. The later 9 Oct report covers a distinct DAC PA4 → ADC PA5 physical experiment.

Physical tracking errors are not structural-parity errors

The ARM 100K final reference/ADC/DAC readings were 2,662 / 2,343 / 2,336 counts; residual to ADC 319 counts. The 9 October RISC-V physical-loop report also retained a nonzero tracking residual. A passing structural gate validates the defined acceptance predicate, not necessarily ideal plant tracking. [S3, S6]

Power snapshots, only at board level

The ARM laboratory supply was observed at 4.98 V and approximately 27–38 mA (about 134–189 mW complete-board input). The later RISC-V bench showed approximately 4.82 V and 10–11 mA (about 48–53 mW complete-board indications). These are setup-specific snapshots, **not isolated OSH-core power or measured energy per decision**. [S3, S6]

09 / Historical latency, instrument boundaries and accuracy

The following are recorded physical measurements under different firmware, peripheral paths and timing boundaries. They are preserved as evidence; they are **not** a controlled three-way speed comparison. [S1, S3, S6]

Campaign / method	Min µs	Mean µs	Max µs	Measurement boundary
ESP32-S3 final 100K	24	29.525	30	Internal final closed-loop route; ADC / OSH / PWM / RC setup.
STM32F446RE 100K	209.500	241.250	241.875	Cortex-M internal cycle counter; declared 16 MHz conversion.
CH32V307 Diligence R1, 9 Oct	281	356.843	361	TIM2 native interval, PA8 marker around ADC-read-to-DAC-write wrapper.

Independent oscilloscope observations

Target	Recorded external interval	Correct interpretation
ESP32-S3	23.0 µs route; 660 ns separate action/selection interval	Two different instrumentation builds and captured windows; neither is a timing distribution.
STM32F446RE	Approximately 226 µs	Externally marked execution interval, not precisely the same boundary as the DWT timing series.
CH32V307, 9 Oct	284 µs external PA8 interval	Single oscilloscope cursor observation of a wrapper-defined ADC-to-DAC window.

No invented latency or accuracy ranking

The retrieved series do not supply P50, P95 or P99 for all three, nor timing observations for the matched 30-case corpus under a common scope. These percentiles cannot be derived from minimum, mean and maximum alone. Likewise, the numeric tolerances in section 02 are thresholds, not measured worst-case deviations, and cannot be interpreted as three-platform ADC error distributions.

A genuine matched latency study would use one ordered case/input sequence, identical initial state, a consistent operation boundary, known clocks, repeated raw cycle counters and explicit decisions on whether ADC acquisition, output writes, UART and settling are included. This document does not claim that a new study of that form occurred.

10 / Integrity anchors: firmware, fixture and captures

Archived artifact	SHA-256 (64 hexadecimal characters)
Compiled 30-case table (4 ELF audit)	`4a77195717d3f87585af84f5e6c0d2d8312ed3cbcb1aaf2f7984ab7405157baf`
Frozen source fixture (3 lineages)	`23863039f2b2b920fe578c6dc6e0e00624ba34e6add00709522be8d6e70ec321`
ESP32-S3 R2 RAM-loaded BIN	`4440dc05a6c4d39dac8b6223feb25303c303fa28b774b6e03d7336ae9aed3811`
ESP32-S3 final physical ELF	`30cbbf62952b442b41aa2bf1398bd9377bd1401a18a5da41a1ea74dd601f1ff3`
ESP32-S3 final physical BIN	`86457e990bd1b96a093f418c5ce8f71b60d41f55c4065757df4ca543c7b4863d`
ARM parity/20K ELF	`9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645`
ARM parity/20K BIN	`336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae`
ARM parity/20K original UART	`c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720`
ARM 100K ELF	`98a420f518d51bbd2d7a797d87df8b2cf825189181d1d7c53ada6ec1d2b95aa1`
ARM 100K BIN	`7ae036ef3983fdbf72081904de591632fd830fc6834ee7aae47ab7d2eb212997`
RISC-V V2 100K ELF	`cb9f88f38e755cb82f51fdfe0bf8000ff0963eb774195e1b3b965ae1975c1517`
RISC-V V2 100K HEX	`544338ec6e5a4fa4251d8dea7f557074418c8014ea297fd2cebaa02f9134ad21`
RISC-V V2 00:16 original UART	`6e01a89c2d17af83c2527e549e142038ae0e66816721b8eaea2b45cc454b7b1d`
RISC-V distinct 9 Oct loop HEX	`2db7f4dba1893c67be1ebcdca9f0b0a6db43484eb9f1fb75e45301bf63bb1268`

These hashes preserve artifact identities; they are not, alone, same-session Flash-readback certificates. The 11:08 RISC-V UART recording has a different digest from the 00:16 recording. No 11:08 digest is substituted here. [S5, S9]

11 / Provenance close-out and evidentiary limits

Link assessed	Disposition	Reason
Frozen 30-case header across lineages	CLOSED FOR ARCHIVED SOURCES	Source SHA-256 identity includes ARM frozen copy reverified on 9 Nov.
30-case reference data inside four archived ELFs	CLOSED IN OWNER-RUN AUDIT	Each matches whole-file reference SHA, and each ELF table equals 49,920 bytes with the same table digest.
ESP32-S3 R2 image → original UART	SUPPORTED BY RAM-LOAD TRANSCRIPT	esptool RAM segment load/execution followed by 30 individually printed PASS cases.
ESP32-S3 R2 ↔ separately archived final ELF	NOT ASSERTED AS BYTE-IDENTICAL	One is the R2 image that printed cases; four-way ELF audit includes the distinct final physical ELF.
ARM 100K programmed image → physical runtime	DOCUMENTED HISTORICALLY	Migration record names 100K ELF and programmer verification followed by run PASS.
RISC-V V2 selected HEX → exact 11:08 image	HISTORICAL PROVENANCE; NO SAME-SESSION FLASH READBACK	Selection and matching archives plus successful UART exist, but exact WCHISPTool verification transcript/readback was not recovered.
Cross-chip, matched per-case latency	NOT TESTED UNDER COMMON BOUNDARY	Three valid but different benchmark methods and windows.

Defensible publication statement

The FARO/OSH native decision path was physically exercised on ESP32-S3, STM32F446RE ARM Cortex-M4 and CH32V307 RISC-V microcontrollers. Each platform's preserved campaign reported successful 30-case structural parity. A later local binary audit identified four historical ELFs containing byte-identical 30-case compiled reference tables. The evidence supports cross-architecture structural decision-reference parity within the tested path, subject to the recorded provenance qualifications. Historical route latencies are available but are not comparable as a single matched ranking.

Beyond the evidence presented

This does not establish a new jointly rerun 90-UART-line experiment, complete same-session Flash-image attestation for every target, a matched per-case speed/accuracy benchmark, independent third-party replication, certified safety, general intelligence, consciousness or portability of every FARO subsystem. These limits do not negate the physical runs that were carried out. [S1–S10]

A / Original ESP32-S3 R2 per-case UART records

The following 30 records are transcribed verbatim from the preserved owner-supplied ESP32-S3 R2 serial transcript. They contain the individual five-level results, number-policy indicator, candidate counts and anonymized meaning hashes. They are **not** ARM or RISC-V UART lines. [S2]

```
CASE=01 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=3 SEL=3 FINAL=2 MEANING=c9bacc8972629cd6
CASE=02 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=03 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=3 SEL=2 FINAL=2 MEANING=c9bacc8972629cd6
CASE=04 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=05 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=06 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=76b81b8fcd10acb8
CASE=07 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=08 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=09 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=beb4b589a11b0162
CASE=10 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=11 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=12 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=d350a6eac0ac33ba
CASE=13 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=14 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=15 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=1785ee2af9510503
CASE=16 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=17 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=18 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=919874dd581c4c43
CASE=19 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=20 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=21 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=4883299398968903
CASE=22 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=23 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=24 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=2 SEL=2 FINAL=2 MEANING=1295c86101e15c45
CASE=25 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=26 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=27 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=fec05c309086dce3
CASE=28 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=CONDITIONED HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=29 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
CASE=30 L1=PASS L2=PASS L3=PASS L4=PASS L5=PASS NUM=STRICT HAL=1 SEL=1 FINAL=1 MEANING=3887d1707a5fdfde
```

Archive interpretation. All cases 01–30 report L1=L2=L3=L4=L5=PASS. The printed 64-bit MEANING field refers to a structural outcome digest, not a language response or an analog actuator action. The original capture also reports its aggregate parity gate PASS.

B / Direct physical gate excerpts and binary comparison

ARM parity/20K capture — original acceptance lines

```
OSH_ARM_EXECUTION_BEGIN
BOARD=NUCLEO-F446RE
CPU=CORTEX-M4
PARITY_CASES=30
PARITY_EXECUTION_BEGIN
PARITY_EXECUTION_END
L1_L2_L3_L4_L5_ALL_30_CASES=PASS
PC_ESP32_PARITY_GATE=PASS
OVERALL_PAPER_CLOSEOUT=PASS
OSH_ARM_EXECUTION_END
```

RISC-V V2 recovery UART — 8 October, 11:08–11:09

```
BOARD=CH32V307VCT6  |  ARCH=RISC-V  |  CPU=QINGKE_V4F
PARITY_CASES=30
PARITY_EXECUTION_BEGIN
PARITY_EXECUTION_END
STRUCTURAL_PARITY_30_30=PASS
GATE_5_OSH_STRUCTURAL_PARITY=PASS
ENDURANCE_CYCLES=100000
RISCV_100K_PROGRESS=100000/100000
STATE_FAILURES=0  |  INVALID_ACTIONS=0
ACTION_INC=174  |  ACTION_HOLD=99667  |  ACTION_DEC=159
RISCV_100K_QUALIFICATION_GATE=PASS
OVERALL_RISCV_V2_100K=PASS
```

9 November local comparison — reported results

Four exact ELF SHA-256 matches; table parser reports 30 cases, 49,920 bytes and identical 64-character table SHA-256 for ARM parity, ARM 100K, RISC-V V2 and ESP32-S3 final. The owner supplied the terminal output; this report did not independently rerun the checker or read firmware from the physical devices.

C / Source register and reproducibility notes

The following documents and records are the basis of the consolidated claims. Titles are descriptive public aliases where archive filenames include private third-party references. Original raw UART and proprietary binaries remain in the owner's custody.

Ref	Source / recorded scope
[S1]	OSH Physical Validation of a Homeostatic Cognitive Architecture, 21 September 2026. ESP32-S3 Step19/Step20, closed-loop test, external oscilloscope, image hashes.
[S2]	Original ESP32-S3 Step19/R2 physical UART transcript, September 2026. RAM-load sequence and 30 individual CASE entries; preserved in owner-provided test logs.
[S3]	ARM Physical Validation, NUCLEO-F446RE, 3 October 2026. Parity, 100K, latencies, oscilloscope, power and artifact hash ledger.
[S4]	ARM Physical Migration Completed, 3 October 2026. STM32CubeProgrammer 100K write/verify/reset and subsequent physical execution.
[S5]	RISC-V Technical Physical Validation Checkpoint, 8 October 2026; plus the later 11:08–11:09 UART recovery record.
[S6]	RISC-V Physical Benchmark and Control Evidence, 9 October 2026. Separate DAC-to-ADC native test, scope interval and board-power observations.
[S7]	FARO 30-case Source Parity Reconstruction, 8 November 2026. Original fixture identity, five-level source predicates, anonymous decision classes.
[S8]	Operator SHA-256 checks, 9 November 2026. ARM frozen fixture and ARM parity UART archive integrity.
[S9]	Owner-run read-only ELF corpus comparison, 9 November 2026. Four whole-file SHA-256 matches; 54 ELF candidates inspected; OSH_PARITY_CASES symbol extraction and four-way 49,920-byte match.
[S10]	Prior consolidated FARO/OSH Physical Parity technical evidence dossier, 9 November 2026; plus the user-supplied Cross-Architecture Physical Execution, Decision Parity and Latency Evidence narrative.

Reproducibility and confidentiality

The public model is evidence-oriented: reviewers can inspect hash anchors, corpus-extraction results, declared acceptance logic, physical-board identity, serial gates and instrumentation records without publication of the proprietary FARO kernel or stored memories. Independent replication requires separately agreed access and independently obtained measurements.

Document generated from owner-supplied historical records and the owner's new technical narrative. It is a consolidation, not a new physical measurement or certification.